Mohamadhosein Sabaghi

Bonn, Germany (Relocating April 2026)
Eligible for 20h/week student employment
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Professional Summary

Backend Software Engineer with 3+ years of experience building scalable distributed systems using NestJS and FastAPI. Strong background in microservices, event-driven architectures (Kafka), and performance optimization. Incoming M.Sc. Computer Science student at University of Bonn seeking a Werkstudent or part-time Software Engineering role starting April 2026.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Backend: NestJS, FastAPI, Django
Databases: PostgreSQL, MongoDB
Messaging & Caching: Kafka, Redis, RabbitMQ
DevOps & Cloud: Docker, Docker Compose, Nginx, AWS S3
Testing: Jest, Pytest
Tools: Git, Swagger, Sentry, Linux, WebSockets

Work Experience

Software Developer — Smartech Sep 2025 – Present

- Developed FastAPI-based backend services supporting 500k+ daily redirect requests.
- Designed and implemented a distributed link shortener service using Kafka and PostgreSQL.
- Contributed to high-throughput messaging systems handling SMS, Email, and Push notifications.
- Reduced average API response time by 20% through Redis caching and query optimization.

Software Developer — TOPS Oct 2023 – Sep 2025

- Designed and maintained backend services using NestJS and Python in monolithic and microservices architectures.
- Implemented Kafka-based asynchronous communication between services to improve scalability.
- Optimized PostgreSQL queries and indexing strategies.
- Integrated Redis caching for high-traffic endpoints.
- Containerized services using Docker and deployed applications using Docker Compose.
- Implemented WebSocket-based real-time features for communication-heavy applications.

- Wrote unit and end-to-end tests (Jest) to ensure reliability.
- Participated in domain modeling and architecture design decisions.

Software Developer — Medic Metaverse (Contract) Aug 2025 – Nov 2025

- Developed backend services using Django within a microservices architecture.
- Integrated PostgreSQL, Redis, Kafka, and object storage systems.
- Contributed to secure handling of sensitive health-related data.

AI Intern — Asr Gooyesh Pardaz Jun 2023 – Sep 2023

- Worked on NLP and text-to-speech systems using Python.
- Trained FastSpeech and VITS models for Persian text-to-speech.
- Developed demo interfaces using Gradio and Telegram bot integrations.

Selected Projects

Scalable Link Shortener Service

- Built a URL shortening service using FastAPI, PostgreSQL, and Kafka.
- Designed database schema for high read/write performance.
- Implemented analytics tracking and caching mechanisms.

Tapsi Garage Web Application

- Developed a backend for multi-role garage management using NestJS and MongoDB.
- Implemented Redis caching and Kafka-based messaging for real-time updates.
- Built automated report generation and secure data handling for internal operations.

Stonade Marketplace Platform

- Developed a bilingual stone marketplace platform with mobile and web admin panels.
- Built backend with NestJS, MongoDB, Redis, and Kafka, supporting multi-role access and real-time chat.
- Deployed using Docker Compose and Nginx; migrated to AWS for international deployment.

Education

M.Sc. Computer Science (Starting April 2026)
University of Bonn, Germany

B.Sc. Computer Engineering (GPA: 18.75/20 — 3.92/4)
Amirkabir University of Technology (Tehran Polytechnic)

Languages

English — Full Professional Proficiency (TOEFL iBT 102)
German — A2 (Currently improving)
Persian — Native